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PRINTING SYSTEM CAPABLE OF SAVING PRINT JOB IN CLOUD STORAGE OTHER THAN STORAGE OF CLOUD PRINT SERVER IN CLOUD PRINT SERVICE, CONTROL METHOD FOR PRINTING SYSTEM, INFORMATION PROCESSING APPARATUS, CONTROL METHOD FOR INFORMATION PROCESSING APPARATUS, AND STORAGE MEDIUM

Abstract

A mechanism capable of saving a print job in a cloud storage other than a storage of a cloud print server in a cloud print service is provided. A printing system providing a cloud print service includes at least one processor, and a memory coupled to the processor storing instructions that, when executed by the processor, cause the processor to perform causing an information processing apparatus, which has received an instruction from a user, to generate a print job and transmit the print job to a set save location, and causing an image forming apparatus, which has received a print execution instruction from the user, to obtain the print job saved in the save location and execute a print processing of the obtained print job. The save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a printing system, a control method for the printing system, an information processing apparatus, a control method for the information processing apparatus, and a storage medium.

Description of the Related Art

[0002] A printing system has been known that provides a cloud print service in which a print job is transmitted from a client terminal, which is an information processing apparatus, to an image forming apparatus via an external cloud server (see Japanese Laid-Open Patent Publication (kokai) No. 2023-85933). In such a printing system, the client terminal obtains document data from a storage location designated by a user, generates a print job based on the obtained document data, and transmits the print job to the image forming apparatus via a cloud print server. Conventionally, in cloud printing, a print job generated by a client terminal

[0003] is saved in a predetermined save location, specifically, in a storage of a cloud print server managed by a printer manufacturer, and then transmitted to an image forming apparatus. For this reason, conventionally, it is not possible to save the print job in a save location other than the storage of the cloud print server, for example, in a cloud storage that is being used by the user.

SUMMARY OF THE INVENTION

[0004] The present invention provides a mechanism capable of saving a print job in a cloud storage other than a storage of a cloud print server in a cloud print service.

[0005] Accordingly, the present invention provides a printing system that

[0006] provides a cloud print service, the printing system comprising at least one processor, and a memory coupled to the processor storing instructions that, when executed by the processor, cause the processor to perform causing an information processing apparatus, which has received a print job transmission instruction from a user, to generate a print job and transmit the print job to a save location that has been set, and causing an image forming apparatus, which has received a print execution instruction from the user, to obtain the print job that has been saved in the save location and execute a print processing of the obtained print job. The save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.

[0007] Accordingly, the present invention provides an information processing apparatus that causes an image forming apparatus, which supports a cloud print service, to execute a print processing, the information processing apparatus comprising at least one processor, and a memory coupled to the processor storing instructions that, when executed by the processor, cause the processor to perform receiving an instruction from a user, and a control of generating a print job in accordance with the instruction and transmitting the print job to a save location that has been set. The image forming apparatus obtains the print job that has been saved in the save location in accordance with a print execution instruction received from the user, and executes a print processing of the obtained print job. The save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.

[0008] According to the present invention, it is possible to save the print job in the cloud storage

other than the storage of the cloud print server in the cloud print service.

[0009] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a configuration diagram that shows an example of a printing

[0011] system according to a first embodiment.

[0012] FIG. 2 is a block diagram that schematically shows a hardware configuration of a client terminal shown in FIG. 1.

[0013] FIG. 3 is a block diagram that schematically shows a hardware configuration of an image forming apparatus shown in FIG. 1.

[0014] FIG. 4A and FIG. 4B are sequence diagrams that show the steps of a first cloud print processing executed by the printing system shown in FIG. 1.

[0015] FIG. 5 is a diagram that shows an example of a setting screen displayed on an output unit shown in FIG. 2.

[0016] FIG. 6 is a diagram that shows an example of a print setting screen displayed on the output unit shown in FIG. 2.

[0017] FIG. 7 is a diagram that shows an example of a cloud print authentication screen displayed on an operation unit shown in FIG. 3.

[0018] FIG. 8 is a configuration diagram that shows an example of a printing system according to a second embodiment.

[0019] FIG. 9A and FIG. 9B are sequence diagrams that show the steps of a second cloud print processing executed by the printing system shown in FIG. 8.

[0020] FIG. 10 is a diagram that shows an example of a cloud storage authentication screen displayed on the operation unit shown in FIG. 3.

[0021] FIG. 11 is a configuration diagram that shows an example of a printing system according to a third embodiment.

[0022] FIG. 12 is a sequence diagram that shows an example of an account linking registration processing executed by the printing system shown in FIG. 11.

[0023] FIG. 13A and FIG. 13B are sequence diagrams that show the steps of a third cloud print processing executed by the printing system shown in FIG. 11.

DESCRIPTION OF THE EMBODIMENTS

[0024] The present invention will now be described in detail below with reference to the accompanying drawings showing embodiments thereof.

[0025] Hereinafter, preferred embodiments of the present invention will be described in detail with reference to the drawings. Here, before describing the configuration of a printing system according to each of the preferred embodiments, the features of the printing system according to each of the preferred embodiments will be described. In the printing system according to each of the preferred embodiments, first, an administrator registers an image forming apparatus in a tenant of a cloud print server. The tenant is a service system deployed on the cloud. Next, the administrator performs settings regarding which users belonging to the tenant are permitted to use the image forming apparatus. A user who has been permitted to use the image forming apparatus submits a print job including document data and a print setting to the cloud print server from a client terminal. The cloud print server saves the received print job in a cloud storage. The image forming apparatus obtains the print job that has been saved in the cloud storage, and executes a print processing of the obtained print job.

[0026] First, a printing system according to a first embodiment of the present invention will be

described.

[0027] FIG. 1 is a configuration diagram that shows an example of a printing system **100** according to the first embodiment.

[0028] As shown in FIG. 1, the printing system **100** includes an image forming apparatus **101**, a cloud storage **106**, a cloud print server **107**, and one or more client terminals that are information processing apparatuses. It should be noted that in the first embodiment, as an example, a configuration in which the printing system **100** includes three client terminals **102** to **104** will be described. In addition, in the first embodiment, a configuration in which the printing system **100** includes one cloud storage **106** will be described as an example, but the present invention is not limited to this configuration. For example, the printing system **100** may be configured to include a plurality of cloud storages. The image forming apparatus **101**, the cloud print server **107**, and the client terminals **102** to **104** are each communicably connected to each other via the Internet.

[0029] The image forming apparatus **101** is an image forming apparatus that supports a cloud print service, and has been registered in the tenant of the cloud print server **107**. The image forming apparatus **101** obtains a print job from the cloud print server **107**, and executes a print processing of the obtained print job.

[0030] The client terminals **102** to **104** each submit a print job to the cloud print server **107**. Each of the client terminals **102** to **104** includes an application **105** capable of saving, editing, and issuing a print instruction for document data.

[0031] The application **105** is, for example, an Internet browser or Adobe Acrobat. In addition, the application **105** is Google Workspace from Google LLC or Microsoft's Office application, which can access and edit document data in the cloud storage **106**. For example, in the client terminal **102**, in accordance with a print instruction received from a user, the application **105** performs a print job generation processing that generates a print job from document data and a print job transmission processing that transmits the print job. These processing are realized by a print framework incorporated in an operating system (an OS) of the client terminal **102**. In the case that the client terminal **102** generates a print job, a printer driver constituting the print framework is used to perform assignment of a print setting and converting document data into a print job.

[0032] It should be noted that in the first embodiment, a configuration in which the client terminals **102** to **104** each include the application **105** with the same functions and configuration will be described, but the present invention is not limited to this configuration. For example, the client terminals **102** to **104** may each include a different application, as long as the application is capable of issuing a print instruction. The user is able to use the print framework from the application **105** to perform the print job generation processing and the print job transmission processing.

[0033] The cloud storage **106** is an online storage service, and provides a service for managing document data and the like online. The cloud storage **106** is, for example, Google Drive from Google LLC or OneDrive from Microsoft.

[0034] The cloud print server **107** is a server managed by the printer manufacturer, and manages print jobs in the cloud print service. For example, the cloud print server **107** is able to save print job related information including a storage location, an access token, etc. of a print job saved in the cloud storage **106** from the client terminal **102** or the like in accordance with an instruction from the user. In addition, the cloud print server **107** is able to transmit the print job related information to the image forming apparatus **101** in response to a request received from the image forming apparatus **101**.

[0035] Next, the hardware configurations of the client terminals **102** to **104** and the cloud print server **107** will be described. It should be noted that in the first embodiment, the client terminals **102** to **104** and the cloud print server **107** have the same hardware configuration, and the hardware configuration of the client terminal **102** will be described below as an example.

[0036] FIG. 2 is a block diagram that schematically shows the hardware configuration of the client terminal **102** shown in FIG. 1. As shown in FIG. 2, the client terminal **102** includes a central

processing unit (a CPU) **201**, a read only memory (a ROM) **202**, a random-access memory (a RAM) **203**, a hard disk drive (an HDD) **204**, an input unit **205**, an output unit **206**, and a communication unit **207**. The CPU **201**, the ROM **202**, the RAM **203**, the HDD **204**, the input unit **205**, the output unit **206**, and the communication unit **207** are connected to each other via a bus **208**.

[0037] The CPU **201** directly or indirectly controls the respective devices connected via the bus **208**, and executes programs for implementing the present invention. The ROM **202** stores a basic input/output system (a BIOS) and the like. The RAM **203** is used as a working area for the CPU **201**, and is also used as a primary storage area for loading software modules for implementing the present invention. The HDD **204** stores the OS, which is the basic software, and various kinds of software modules. It should be noted that in the first embodiment, the client terminal **102** may be configured to include a storage device such as a solid state drive (an SSD) that has the same functions as an HDD, instead of an HDD. The input unit **205** includes a keyboard, a pointing device, and the like. The output unit **206** is, for example, a display that displays various kinds of information. The communication unit **207** is an interface for connecting to a network.

[0038] FIG. **3** is a block diagram that schematically shows a hardware configuration of the image forming apparatus **101** shown in FIG. **1**. As shown in

[0039] FIG. **3**, the image forming apparatus **101** includes a CPU **301**, a ROM **302**, a RAM **303**, an HDD **304**, an operation unit **305**, an input unit **306**, a printer **307**, and a communication unit **308**. The CPU **301**, the ROM **302**, the RAM **303**, the HDD **304**, the operation unit **305**, the input unit **306**, the printer **307**, and the communication unit **308** are connected to each other via a bus **309**.

[0040] The CPU **301** directly or indirectly controls the respective devices connected via the bus **309**, and executes programs for implementing the present invention. The ROM **302** stores a BIOS and the like. The RAM **303** is used as a working area for the CPU **301**, and is also used as a primary storage area for loading software modules for implementing the present invention. The HDD **304** stores an OS, which is the basic software, and various kinds of software modules. It should be noted that in the first embodiment, the image forming apparatus **101** may be configured to include an SSD or the like that has the same functions as an HDD, instead of an HDD. The operation unit **305** is a display that is capable of being operated by touch. The input unit **306** includes a numeric keypad, an IC card reader, and the like. The printer **307** is a printer engine that executes a print processing of a print job received from the outside via the communication unit **308**. The communication unit **308** is a network interface for connecting to the Internet or an office LAN.

[0041] FIG. **4A** and FIG. **4B** are sequence diagrams that show the steps of a first cloud print processing executed by the printing system **100** shown in FIG. **1**. In FIG. **4A** and FIG. **4B**, as an example, a configuration in which the first cloud print processing is performed by the client terminal **102**, the cloud print server **107**, the cloud storage **106**, and the image forming apparatus **101** will be described. In FIG. **4A** and FIG. **4B**, the processing by the client terminal **102** is realized by the CPU **201** loading a program stored in the ROM **202** or the like into the RAM **203** and executing the program. The processing by the cloud print server **107** is realized by a CPU of the cloud print server **107** loading a program stored in a ROM or the like of the cloud print server **107** into a RAM of the cloud print server **107** and executing the program. The processing by the image forming apparatus **101** is realized by the CPU **301** loading a program stored in the ROM **302** or the like into the RAM **303** and executing the program.

[0042] It should be noted that in FIG. **4A** and FIG. **4B**, it is assumed that in the client terminal **102**, authentication information for the cloud storage **106** and authentication information for the cloud print server **107** have been outputted from the application **105** to the print framework. The authentication information for the cloud storage **106** is authentication information used when logging in to the cloud storage **106**, and is, for example, a username and a password. The authentication information for the cloud print server **107** is authentication information used when logging in to the cloud print server **107**, and is, for example, a username and a password. When the

print framework logs in with the authentication information, the print framework is able to perform transmitting a print job to the cloud storage **106** and transmitting print job related information to the cloud print server **107**. In addition, the application **105** is able to download document data saved in the cloud storage **106**, and edit or view document data saved in the local environment or the cloud storage **106**. The application **105** is also able to save the print job in the cloud storage **106** from the print framework.

[0043] It should be noted that in the first embodiment, it is assumed that the authentication information for the cloud storage **106** has been set in advance by the user on a setting screen **500** of FIG. 5 displayed on the output unit **206** of the client terminal **102** before the first cloud print processing is performed.

[0044] FIG. 5 is a diagram that shows an example of the setting screen **500** displayed on the output unit **206** shown in FIG. 2. The setting screen **500** includes a combo box **501**, a save destination folder name setting field **502**, a username setting field **503**, a password setting field **504**, a check box **505**, an OK button **506**, and a cancel button **507**. The combo box **501** is a combo box for designating a cloud storage. For example, in the case that the printing system **100** includes a plurality of cloud storages, the combo box **501** allows one cloud storage to be selected from the plurality of cloud storages. A directory for saving a print job in the cloud storage that has been selected by the combo box **501** is inputted into the save destination folder name setting field **502**. This directory is saved as storage destination information for the print job. Authentication information for the cloud storage that has been selected by the combo box **501**, specifically, a username to be used when connecting to the cloud storage that has been selected by the combo box **501**, is inputted into the username setting field **503**. The authentication information for the cloud storage that has been selected by the combo box **501**, specifically, a password to be used when connecting to the cloud storage that has been selected by the combo box **501**, is inputted into the password setting field **504**. The check box **505** is a check box for setting whether or not to require input of a username and a password each time when connecting to the cloud storage that has been selected by the combo box **501**, without saving the information in the username setting field **503** and the information in the password setting field **504**. When the check box **505** is selected, the setting values that have been set by the username setting field **503** and the password setting field **504** are cleared, and input operations in the username setting field **503** and the password setting field **504** are disabled. When the OK button **506** is selected, the setting values that have been set on the setting screen **500** are saved in the HDD **204** of the client terminal **102** as the authentication information for the cloud storage that has been selected by the combo box **501**, and the setting screen **500** is closed. When the cancel button **507** is selected, the setting values that have been set on the setting screen **500** are not saved in the HDD **204** of the client terminal **102**, and the setting screen **500** is closed.

[0045] As shown in FIG. 4A and FIG. 4B, first, in **S401** (step **401**), the user issues a print instruction of document data that he or she has been editing or viewing by using the application **105** of the client terminal **102**.

[0046] Next, in **S402**, the application **105** transmits, to the CPU **201** of the client terminal **102**, the document data and a print start request of the document data.

[0047] Next, in **S403**, the client terminal **102** displays a print setting screen shown in FIG. 6 on the output unit **206**.

[0048] FIG. 6 is a diagram that shows an example of a print setting screen **600** displayed on the output unit **206** shown in FIG. 2. The print setting screen **600** is a screen for performing a print setting of the obtained document data and setting of a save location of a print job. The print setting screen **600** includes tabs **601** to **604**, a combo box **605**, a login setting button **606**, a save destination folder name setting field **607**, a username setting field **608**, a password setting field **609**, a check box **610**, and a connection confirmation button **611**. In addition, the print setting screen **600** includes an OK button **612** and a cancel button **613**.

[0049] The tabs **601** to **604** are tab buttons for switching to corresponding setting screens. For example, when the tab **601** is selected, a screen for performing a basic print setting is displayed. When the tab **602** is selected, a screen for performing a print setting for the page is displayed. When the tab **603** is selected, a screen for setting the print quality is displayed. In FIG. 6, as an example, the tab **604** is selected, and a screen for performing the setting of the save location of the print job is displayed.

[0050] The combo box **605** is a combo box for designating the save location of the print job. By using the combo box **605**, it is possible to designate a cloud storage other than a storage of the cloud print server **107**. For example, in the case that the printing system **100** includes a plurality of cloud storages, the combo box **605** allows one cloud storage to be selected from the plurality of cloud storages. It should be noted that in the first embodiment, a configuration, in which for example, only cloud storage(s) that the image forming apparatus **101** is able to access from among the plurality of cloud storages included in the printing system **100** is/are displayed on the combo box **605**, may be adopted. In addition, in the case that the settings on the setting screen **500** have been performed in advance, a configuration, in which the setting values that have been set in the items **502** to **505** on the setting screen **500** are automatically set in the items **607** to **610**, respectively, may be adopted.

[0051] The login setting button **606** is a button for opening the setting screen **500**. A directory for saving a print job in the cloud storage that has been selected by the combo box **605** is inputted into the save destination folder name setting field **607**. This directory is saved as storage destination information for the print job. A username as authentication information to be used when connecting to the cloud storage that has been selected by the combo box **605** is inputted into the username setting field **608**. A password as the authentication information to be used when connecting to the cloud storage that has been selected by the combo box **605** is inputted into the password setting field **609**. The check box **610** is a check box for saving the information in the username setting field **608** and the information in the password setting field **609**, and for making the input of the next time unnecessary.

[0052] The connection confirmation button **611** is a button for confirming whether or not it is possible to connect to the cloud storage that has been selected by the combo box **605**. When the OK button **612** is selected, the setting values that have been set on the print setting screen **600** are saved in the RAM **203** of the client terminal **102**, and the print setting screen **600** is closed. At this time, in the case that the check box **610** has been selected, the setting values that have been set on the print setting screen **600** are saved in the HDD **204** of the client terminal **102** as the authentication information for the cloud storage that has been selected by the combo box **605**, and the print setting screen **600** is closed. When the cancel button **613** is selected, the setting values that have been set on the print setting screen **600** are not saved in the RAM **203** of the client terminal **102**, and the print setting screen **600** is closed.

[0053] Returning to FIG. 4A and FIG. 4B, in **S404**, the user inputs the print setting and a transmission destination setting of the document data on the print setting screen **600** displayed on the output unit **206**. In the transmission destination setting, on the screen displayed by selecting the tab **604**, a cloud storage that becomes the save location of the print job is designated. In the first embodiment, it is assumed that the cloud storage **106** has been designated.

[0054] Next, in **S405**, the user confirms that the setting values that have been inputted into the print setting screen **600** displayed on the output unit **206** are correct, and issues a print job transmission instruction to the client terminal **102** (instructs the client terminal **102** to transmit a print job).

[0055] Next, in **S406**, the client terminal **102** performs a print job generation processing. In the print job generation processing, a print job including the document data and the print setting that has been inputted into the print setting screen **600** is generated.

[0056] Next, in **S407**, the client terminal **102** obtains the authentication information for the cloud storage **106** from the HDD **204**. In addition, the client terminal **102** obtains the storage destination

information from the HDD **204**.

[0057] Next, in **S408**, the client terminal **102** performs a login processing with respect to the cloud storage **106** by using the obtained authentication information. It should be noted that in the case that the authentication information for the cloud storage **106** has not been saved in the HDD **204**, in **S407**, the client terminal **102** displays, on the output unit **206**, a dialog box that prompts the user to input a username and a password. Furthermore, the client terminal **102** performs the login processing of **S408** by using the username and the password that have been inputted.

[0058] In **S409**, the cloud storage **106** transmits a login processing result notification to the client terminal **102**.

[0059] The client terminal **102**, which has received a login processing result notification indicating the success of the login processing from the cloud storage **106**, transmits in **S410** the print job, the storage destination information, and a print job save request to the cloud storage **106**. It should be noted that in the case that the client terminal **102** has received a login processing result notification indicating the failure of the login processing, the client terminal **102** may display, on the output unit **206**, a dialog box that prompts the user to input a username and a password, and perform the login processing again by using the username and the password that have been inputted.

[0060] Next, in **S411**, the cloud storage **106** performs a file save processing that saves the print job in the directory indicated by the storage destination information. Next, in **S412**, the cloud storage **106** transmits, to the client terminal **102**, a save completion notification which indicates that saving of the print job has been completed. It should be noted that in the case that the remaining capacity of the cloud storage **106** is equal to or less than the data amount of the print job that has been transmitted from the client terminal **102** in **S410**, the cloud storage **106** is not able to store this print job. In this case, in **S412**, the cloud storage **106** transmits a save failure notification to the client terminal **102**. The client terminal **102** that has received the save failure notification may notify the user that saving of the print job has failed, and may return the processing to **S407**, prompting the user to select another cloud storage.

[0061] Next, in **S413**, the client terminal **102** transmits, to the cloud storage **106**, a request for issuing access information used when the image forming apparatus **101** obtains the print job that has been saved in the cloud storage **106**, for example, a request for issuing an access token for the print job.

[0062] Next, the cloud storage **106** issues an access token with an expiration date, and in **S414**, transmits this access token to the client terminal **102**.

[0063] Next, the client terminal **102** generates print job related information. The print job related information includes information indicating the print job, such as the name of the print job transmitted in **S410**, rather than the entity data of the print job. Furthermore, the print job related information includes the storage destination information transmitted in **S410**, information indicating the cloud storage **106** that is the transmission destination of the storage destination information, the access token obtained from the cloud storage **106**, information about the expiration date of this access token, etc. In other words, the print job related information does not include the entity data of the print job. In **S415**, the client terminal **102** transmits this print job related information to the cloud print server **107**.

[0064] Next, in **S416**, the cloud print server **107** performs a processing for updating a print job list that manages the print jobs that have been saved in the cloud storage **106** (an update processing of the print job list). In this update processing, information about the print job that has been newly saved in the cloud storage **106** in **S411** is added to the print job list in which information about the print jobs that have been saved in the cloud storage **106** up until now has been registered. Next, in **S417**, the cloud print server **107** transmits a print job transmission completion notification to the client terminal **102**.

[0065] The steps of **S401** to **S417** are the description of the procedure for saving the print job in the cloud storage **106** and for the cloud print server **107** to register the information about the print job

in the print job list.

[0066] Next, the flow, in which the image forming apparatus **101** obtains the print job related information from the cloud print server **107**, obtains a print job corresponding to the print job related information from the cloud storage **106**, and executes a print processing, will be described.

[0067] In **S418**, the user instructs any image forming apparatus that is able to access the cloud print server **107** to execute cloud print. In the first embodiment, as an example, it is assumed that the user has instructed the image forming apparatus **101** to execute cloud print by operating the operation unit **305** of the image forming apparatus **101**.

[0068] Next, in **S419**, the image forming apparatus **101** causes the operation unit **305** to display a cloud print authentication screen **700** of FIG. 7 which prompts the user to input authentication information required to access the cloud print server **107**.

[0069] FIG. 7 is a diagram that shows an example of the cloud print authentication screen **700** displayed on the operation unit **305** shown in FIG. 3. The cloud print authentication screen **700** includes a username input field **701**, a password input field **702**, a login button **703**, and a “login with IC card” button **704**.

[0070] A username as authentication information to be used when connecting to the cloud print server **107** is inputted into the username input field **701**. A password as the authentication information to be used when connecting to the cloud print server **107** is inputted into the password input field **702**. When the login button **703** is selected, the image forming apparatus **101** transmits the authentication information (the username and the password) inputted by the cloud print authentication screen **700** to the cloud print server **107**, and closes the cloud print authentication screen **700**. The cloud print server **107** executes a login processing by using the received authentication information.

[0071] The “log in with IC card” button **704** is displayed in the case that the image forming apparatus **101** includes an IC card reader or the like in the input unit **306**. When the user selects the “log in with IC card” button **704** and touches the input unit **306** with an IC card, the image forming apparatus **101** reads authentication information from the IC card. The image forming apparatus **101** transmits this authentication information to the cloud print server **107**, and closes the cloud print authentication screen **700**. The cloud print server **107** executes a login processing by using the received authentication information.

[0072] Returning to FIG. 4A and FIG. 4B, in **S420**, the user inputs the authentication information for the cloud print server **107** into the image forming apparatus **101**. In the first embodiment, the authentication information for the cloud print server **107** is inputted into the image forming apparatus **101** when the user directly inputs the authentication information into the operation unit **305** or when the user touches the input unit **306** with the IC card. Next, in **S421**, the image forming apparatus **101** transmits the inputted authentication information to the cloud print server **107**.

[0073] Next, in **S422**, the cloud print server **107** performs an authentication processing by using the obtained authentication information. In **S423**, the cloud print server **107** that has been successfully authenticated transmits, to the image forming apparatus **101**, an authentication completion notification and print job list information that is obtained from the print job list. The print job list information is list information of the print jobs that have been saved in the cloud storage **106**.

[0074] Next, in **S424**, the image forming apparatus **101** causes the operation unit **305** to display a print job list screen (not shown) by using the received print job list information. Next, in **S425**, the user selects a print job to be printed on the print job list screen. Next, in **S426**, the image forming apparatus **101** reflects on the print job list screen that the print job has been selected by the user (the print job is in a selected state).

[0075] Next, in **S427**, the user issues a print execution instruction of the selected print job to the image forming apparatus **101**. Next, in **S428**, the image forming apparatus **101** transmits, to the cloud print server **107**, a request for obtaining print job related information corresponding to the print job that has been selected by the user in **S425** (an obtaining request). Next, in **S429**, the cloud

print server **107** transmits, to the image forming apparatus **101**, print job related information corresponding to the obtaining request that has been received from the image forming apparatus **101**.

[0076] Next, in **S430**, the image forming apparatus **101** transmits, to the cloud storage **106**, the access token included in the received print job related information and a request for obtaining a print job corresponding to this print job related information (a print job obtaining request). It should be noted that in **S430**, the transmission destination of the access token and the print job obtaining request is identified as the cloud storage **106** based on the storage destination information and the like that are included in this print job related information.

[0077] Next, in **S431**, the cloud storage **106** transmits, to the image forming apparatus **101**, a print job corresponding to the received print job obtaining request. Next, in **S432**, the image forming apparatus **101** executes a print processing of the received print job, and ends the first cloud print processing.

[0078] As described above, in the above-described embodiment, unlike the conventional method of saving a print job in the storage of the cloud print server **107**, a print job is saved in the cloud storage **106** that is designated by the user. As a result, in the cloud print service, it is possible to save a print job in the cloud storage other than the storage of the cloud print server **107**. As a result, a user who feels that there is a risk in saving a print job in the storage of the cloud print server **107** from the viewpoint of security or the like is able to feel at ease by being allowed to use the cloud storage **106** which he or she is familiar with. In addition, with the above-described configuration, the cloud print server **107** does not save print jobs with a relatively large amount of data, so the cloud print server **107** does not need to be equipped with a large-capacity storage. In addition, since the cloud print server **107** does not transmit or receive the entity data of a print job, which has a relatively large amount of data, the communication load of the cloud print server **107** is reduced. As a result, it is also possible to reduce the management costs of the cloud print server **107** at the printer manufacturer.

[0079] In addition, in the above-described embodiment, the save location of the print job is a storage that has been selected by the user from among the plurality of cloud storages. As a result, in the configuration in which the user uses the plurality of cloud storages, the user is able to save a print job in the cloud storage of his or her choice.

[0080] Next, a printing system according to a second embodiment of the present invention will be described.

[0081] The second embodiment is basically the same as the first embodiment described above in terms of configuration and operation, but differs from the first embodiment described above in that the printing system does not include a cloud print server. Therefore, descriptions of overlapping configurations and operations will be omitted, and only different configurations and operations will be described below.

[0082] Here, as a cloud print system that does not use a cloud print server, a configuration, in which an image forming apparatus directly obtains document data that has been stored in a cloud storage being used by a user of OneDrive, Google Drive, or the like, has been known. In such a configuration, since there is no need to sign a contract for a cloud print service that uses a cloud print server, it is possible to reduce the cost of using cloud print. On the other hand, in such a configuration, since the cloud storage saves document data before it is converted into a print job, detailed print settings that are capable of being set by the printer driver or the like, and settings of functions unique to the image forming apparatus cannot be made. Furthermore, there is a problem that only document data in a format that is capable of being directly processed by the image forming apparatus can be printed.

[0083] In order to deal with this issue, in the second embodiment, the print job converted from the document data is saved in the cloud storage **106** designated by the user.

[0084] FIG. **8** is a configuration diagram that shows an example of a printing system **800** according

to the second embodiment. The printing system **800** is different from the above-described printing system **100** in that the printing system **800** does not include the cloud print server **107**. Specifically, the printing system **800** includes the image forming apparatus **101**, the cloud storage **106**, and one or more client terminals. It should be noted that in the second embodiment, as an example, a configuration in which the printing system **800** includes three client terminals **102** to **104** will be described. In addition, in the second embodiment, a configuration in which the printing system **800** includes one cloud storage **106** will be described as an example, but the present invention is not limited to this configuration. For example, the printing system **800** may be configured to include a plurality of cloud storages. The image forming apparatus **101**, the cloud print server **107**, and the client terminals **102** to **104** are each communicably connected to each other via the Internet. In the printing system **800**, the client terminal **102** transmits a print job to the cloud storage **106**, and the image forming apparatus **101** obtains the print job from the cloud storage **106** and performs a print processing.

[0085] FIG. **9A** and FIG. **9B** are sequence diagrams that show the steps of a second cloud print processing executed by the printing system **800** shown in FIG. **8**. In FIG. **9A** and FIG. **9B**, as an example, a configuration in which the second cloud print processing is performed by the client terminal **102**, the cloud storage **106**, and the image forming apparatus **101** will be described. In FIG. **9A** and FIG. **9B**, the processing by the client terminal **102** is realized by the CPU **201** loading a program stored in the ROM **202** or the like into the RAM **203** and executing the program. The processing by the image forming apparatus **101** is realized by the CPU **301** loading a program stored in the ROM **302** or the like into the RAM **303** and executing the program. It should be noted that the second cloud print processing is a processing similar to the first cloud print processing described above, and the following will particularly describe the differences from the first cloud print processing described above. In the second cloud print processing, similar to the first cloud print processing, it is assumed that the settings on the setting screen **500** described above have been performed in advance.

[0086] As shown in FIG. **9A** and FIG. **9B**, first, the processes of **S901** to **S912**, which are the same as the above-described processes of **S401** to **S412**, are performed. The steps of **S901** to **S912** are the description of the procedure for saving the print job in the cloud storage **106** from the client terminal **102**.

[0087] Next, the flow, in which the image forming apparatus **101** obtains a print job from the cloud storage **106** and executes a print processing, will be described.

[0088] In **S913**, the user instructs any image forming apparatus that is able to access the cloud storage **106** to execute cloud print. In the second embodiment, as an example, it is assumed that the user has instructed the image forming apparatus **101** to execute cloud print by operating the operation unit **305** of the image forming apparatus **101**.

[0089] Next, in **S914**, the image forming apparatus **101** causes the operation unit **305** to display a cloud storage authentication screen **1000** of FIG. **10** which prompts the user to input authentication information required to access the cloud storage **106**.

[0090] FIG. **10** is a diagram that shows an example of the cloud storage authentication screen **1000** displayed on the operation unit **305** shown in FIG. **3**.

[0091] The cloud storage authentication screen **1000** includes a combo box **1001**, a username input field **1002**, a password input field **1003**, a default folder name setting field **1004**, a check box **1005**, and a login button **1006**.

[0092] The combo box **1001** is a combo box for designating a cloud storage to be used. By using the combo box **1001**, it is possible to designate a cloud storage other than the storage of the cloud print server **107**. For example, in the case that the printing system **800** includes a plurality of cloud storages, the combo box **1001** allows one cloud storage to be selected from the plurality of cloud storages. It should be noted that a configuration, in which for example, only cloud storage(s) that the image forming apparatus **101** is able to access from among the plurality of cloud storages

included in the printing system **800** is/are displayed on the combo box **1001**, may be adopted. A username as authentication information to be used when connecting to the cloud storage that has been selected by the combo box **1001** is inputted into the username input field **1002**. A password as the authentication information to be used when connecting to the cloud storage that has been selected by the combo box **1001** is inputted into the password input field **1003**. A directory for saving a print job in the cloud storage that has been selected by the combo box **1001** is inputted into the default folder name setting field **1004**. This directory is designated as a default directory, and is saved as storage destination information for the print job.

[0093] The check box **1005** is a check box for selecting whether or not to use the default directory that has been set by the default folder name setting field **1004**.

[0094] In the case that the check box **1005** has been selected, when executing cloud printing, the image forming apparatus **101** obtains the print job from the default directory. On the other hand, in the case that the check box **1005** has not been selected, the default folder name setting field **1004** is deleted, and when executing cloud printing, the user needs to designate the directory in which the print job has been saved. The login button **1006** is a button for issuing an authentication instruction of the selected cloud storage.

[0095] Returning to FIG. 9A and FIG. 9B, in S915, by means of the cloud storage authentication screen **1000**, the user selects a cloud storage to be used and inputs authentication information to be used when connecting to the selected cloud storage. In the following, as an example, it is assumed that the user has selected the cloud storage **106** as the cloud storage to be used. In addition, the user selects the login button **1006** to issue an authentication instruction of the cloud storage **106**. When the login button **1006** is selected, the setting values that have been set on the cloud storage authentication screen **1000** are transmitted to the cloud storage **106**, and the cloud storage authentication screen **1000** is closed.

[0096] It should be noted that in the second embodiment, the user needs to select the cloud storage (perform a cloud storage selection operation) on both the client terminal **102** (for example, in S904) and the image forming apparatus **101** (for example, in S915). Therefore, a configuration may be adopted in which it is possible to set in advance so that the same cloud storage is displayed in both the combo box **605** of the print setting screen **600** and the combo box **1001** of the cloud storage authentication screen **1000** in a state where the same cloud storage is designated as the initial value. With such a configuration, it is possible to reduce the labor and time required for performing the cloud storage selection operation.

[0097] Next, in S916, the image forming apparatus **101** performs a login processing to the cloud storage **106** by using the authentication information that has been inputted by means of the cloud storage authentication screen **1000**.

[0098] Next, in S917, the cloud storage **106** transmits a login processing result notification to the client terminal **102**.

[0099] The image forming apparatus **101**, which has received a login processing result notification indicating the success of the login processing from the cloud storage **106**, transmits to the cloud storage **106** in S918 a request for obtaining directory information indicating the directory structure of the cloud storage **106** (a directory information obtaining request). It should be noted that in the case that the image forming apparatus **101** has received a login processing result notification indicating the failure of the login processing, the image forming apparatus **101** may return to S914 and request the user to input the authentication information again.

[0100] Upon receiving the directory information obtaining request, the cloud storage **106** transmits in S919 the directory information of the cloud storage **106** to the image forming apparatus **101**.

[0101] Next, in S920, the image forming apparatus **101** causes the operation unit **305** to display a storage destination directory selection screen (not shown) by using the obtained directory information. The storage destination directory selection screen is a screen that prompts the user to select a directory in which a print job to be printed has been saved.

[0102] Next, in **S921**, the user selects a directory in which a print job to be printed has been saved by means of the storage destination directory selection screen.

[0103] Next, in **S922**, the image forming apparatus **101** transmits a request for obtaining storage destination directory information. The storage destination directory information is list information of print jobs that have been stored in the directory selected by the user in **S921**.

[0104] Next, in **S923**, the cloud storage **106** transmits storage destination directory information to the image forming apparatus **101**. It should be noted that in the case that the check box **1005** has been selected on the cloud storage authentication screen **1000** in **S915**, after the process of **S919** is performed, the processes of **S920** to **S923** may not be performed and the process of **S924** may be performed.

[0105] In **S924**, the image forming apparatus **101** displays a print job list screen by using the storage destination directory information. After that, the processes of **S925** to **S927**, which are the same as the above-described processes of **S425** to **S427**, are performed.

[0106] Next, in **S928**, the image forming apparatus **101** transmits, to the cloud storage **106**, a request for obtaining the print job that has been selected by the user on the print job list screen (an obtaining request). Next, in **S929**, the cloud storage **106** transmits, to the image forming apparatus **101**, a print job corresponding to the received obtaining request. Next, in **S930**, the image forming apparatus **101** executes a print processing of the received print job. Thereafter, the second cloud print processing ends.

[0107] In the second embodiment, the client terminal **102** transmits the print job including the document data and the print setting to the cloud storage **106**, and the image forming apparatus **101** obtains the print job from the cloud storage **106** and executes the print processing. That is, the cloud print server **107** is not used. As a result, it is possible to realize cloud print that performs the print processing of the print job including the print setting set on the print setting screen **600** while suppressing the usage cost of the cloud print server **107**.

[0108] It should be noted that in the second embodiment, the image forming apparatus **101** may be registered in advance in the client terminal **102** so that a print job optimal for the image forming apparatus **101** is generated.

[0109] In addition, in the above-described embodiment, the configuration in which the print job is stored in the cloud storage **106** has been described, but the present invention is not limited to this configuration. For example, the print job may be stored in a storage of the client terminal **102** such as the HDD **204**, or the print job may be stored in a storage or the like of the client terminals **103** and **104** other than the client terminal **102**.

[0110] Next, a printing system according to a third embodiment of the present invention will be described.

[0111] The third embodiment is basically the same as the first embodiment described above in terms of configuration and operation, but differs from the first embodiment described above in that an account for the cloud print server **107** is linked to an account for the cloud storage **106**.

Therefore, descriptions of overlapping configurations and operations will be omitted, and only different configurations and operations will be described below.

[0112] FIG. **11** is a configuration diagram that shows an example of a printing system **1100** according to the third embodiment. The printing system **1100** is different from the above-described printing system **100** shown in FIG. **1** in that the cloud storage **106** is connected to the cloud print server **107** via the Internet. The printing system **1100** includes the image forming apparatus **101**, the cloud storage **106**, the cloud print server **107**, and one or more client terminals. It should be noted that in the third embodiment, as an example, a configuration in which the printing system **1100** includes three client terminals **102** to **104** will be described. In addition, in the third embodiment, a configuration in which the printing system **1100** includes one cloud storage **106** will be described as an example, but the present invention is not limited to this configuration. For example, the printing system **1100** may be configured to include a plurality of cloud storages. The

image forming apparatus **101**, the cloud print server **107**, and the client terminals **102** to **104** are each communicably connected to each other via the Internet.

[0113] FIG. **12** is a sequence diagram that shows an example of an account linking registration processing executed by the printing system **1100** shown in FIG. **11**.

[0114] The account linking registration processing is a processing in which an administrator having administrator authority of the cloud print server **107** registers an account of a general user of the cloud print server **107** and an account of the general user of the cloud storage **106** in association with each other (in linking to each other). It should be noted that the general user of the cloud print server **107** is a user who does not have the administrator authority of the cloud print server **107**. The account of the general user of the cloud print server **107** is an account for the general user to use the cloud print server **107**. The account of the general user of the cloud storage **106** is an account for the general user to use the cloud storage **106**.

[0115] The account linking registration processing is performed by the client terminal **104** that is operated by the administrator, the cloud storage **106**, and the cloud print server **107**. In the account linking registration processing, the processing by the client terminal **104** is realized by a CPU of the client terminal **104** loading a program stored in a ROM or the like of the client terminal **104** into a RAM of the client terminal **104** and executing the program. The processing by the cloud print server **107** is realized by the CPU of the cloud print server **107** loading a program stored in the ROM or the like of the cloud print server **107** into the RAM of the cloud print server **107** and executing the program.

[0116] As shown in FIG. **12**, in **S1201**, the client terminal **104** executes a login processing to the cloud print server **107** with an account having the administrator authority of the cloud print server **107**.

[0117] In **S1202**, the client terminal **104** that has logged in with the account having the administrator authority of the cloud print server **107** transmits, to the cloud print server **107**, an instruction for registering account information of a general user of the cloud print server **107** (a registration instruction). This registration instruction includes the account information of the general user to be registered, for example, includes an account name of this general user and authentication information thereof.

[0118] Next, in **S1203**, the cloud print server **107** performs a registration processing of the account information of the general user (a processing that registers the account information of the general user) in accordance with the accepted registration instruction. This allows the cloud print server **107** to be used with the registered account information. It should be noted that in the case that the account information of the general user has already been registered, the processes of **S1202** and **S1203** may be omitted.

[0119] Next, in **S1204**, the client terminal **104** transmits a cloud storage account linking instruction to the cloud print server **107**. The cloud storage account linking instruction includes the account information of the general user, and storage destination information indicating a directory of the cloud storage **106** in which a print job of the general user is to be saved.

[0120] Next, in **S1205**, the cloud print server **107** executes an account linking processing. In the account linking processing, the account of the general user of the cloud print server **107** is linked to the account of the general user of the cloud storage **106** indicated by the storage destination information. In addition, the storage destination information is saved in an HDD of the cloud print server **107** in association with these accounts (in linking to these accounts).

[0121] Next, in **S1206**, the cloud print server **107** executes a login processing to the cloud storage **106** by using the account of the cloud storage **106** that has been linked to the account of the general user of the cloud print server **107**.

[0122] In **S1207**, the cloud storage **106** that has been successfully logged in transmits a login completion notification to the cloud print server **107**.

[0123] Next, in **S1208**, the cloud print server **107** transmits, to the client terminal **104**, a user

registration completion notification indicating that the account linking registration processing has been successful. Thereafter, the account linking registration processing ends.

[0124] It should be noted that the account of the general user of the cloud print server **107** may be linked to a plurality of accounts of the cloud storage **106**. In addition, the priority order of the cloud storage to be used, and the job save capacity limit for the storage destination of each print job may be set.

[0125] In addition, when the general user logs in to the cloud print server **107** for the first time, the general user may be able to reset the authentication information and the storage destination of the print job in the cloud storage and the cloud print server **107** to which the account of the general user has been linked.

[0126] In addition, an expiration date may be set for each type of authentication information, and the general user may be notified of the expiration date of the authentication information by being prompted to reset the authentication information a certain period of time (for example, ten days) before the expiration date.

[0127] FIG. **13A** and FIG. **13B** are sequence diagrams that show the steps of a third cloud print processing executed by the printing system **1100** shown in FIG. **11**. In FIG. **13A** and FIG. **13B**, as an example, a configuration in which the third cloud print processing is performed by the client terminal **102**, the cloud print server **107**, the cloud storage **106**, and the image forming apparatus **101** will be described. It should be noted that the client terminal **102** is a terminal operated by a general user (hereinafter, simply referred to as “a user”). In FIG. **13A** and FIG. **13B**, the processing by the client terminal **102** is realized by the CPU **201** loading a program stored in the ROM **202** or the like into the RAM **203** and executing the program. The processing by the cloud print server **107** is realized by the CPU of the cloud print server **107** loading a program stored in the ROM or the like of the cloud print server **107** into the RAM of the cloud print server **107** and executing the program. The processing by the image forming apparatus **101** is realized by the CPU **301** loading a program stored in the ROM **302** or the like into the RAM **303** and executing the program. It should be noted that the third cloud print processing is a processing similar to the first cloud print processing described above, and the following will particularly describe the differences from the first cloud print processing described above. In FIG. **13A** and FIG. **13B**, it is assumed that the account linking registration processing shown in FIG. **12** described above has been executed, and an account of a user of the cloud print server **107** and an account of the user of the cloud storage **106** have been linked to each other and registered. In addition, it is assumed that in the client terminal **102**, the authentication information for the cloud storage **106** and the authentication information for the cloud print server **107** have been outputted from the application **105** to the print framework.

[0128] As shown in FIG. **13A** and FIG. **13B**, first, the processes of **S1301** to **S1302**, which are the same as the above-described processes of **S401** to **S402**, are performed.

[0129] Next, in **S1303**, the client terminal **102** causes the output unit **206** to display a print setting screen. This print setting screen differs from the print setting screen **600** shown in FIG. **6** described above in that this print setting screen does not include the tab **604** for switching to a screen for performing the setting of the cloud storage. Therefore, in the printing system **1100**, the user does not need to select the cloud storage on the print setting screen (perform the cloud storage selection operation on the print setting screen).

[0130] Next, in **S1304**, the user inputs a print setting of the document data on the print setting screen displayed on the output unit **206**. Next, in **S1305**, the user confirms that the setting values that have been inputted into the print setting screen displayed on the output unit **206** are correct, and issues a print job transmission instruction to the client terminal **102** (instructs the client terminal **102** to transmit a print job).

[0131] Next, in **S1306**, the client terminal **102** performs a print job generation processing. In the print job generation processing, a print job including the document data and the print setting that

has been inputted into the print setting screen is generated. Next, in **S1307**, the client terminal **102** transmits the print job to the cloud print server **107**.

[0132] Next, in **S1308**, the cloud print server **107** obtains authentication information for the account of the user of the cloud storage **106**, which has been linked to the account of the user of the cloud print server **107**, and storage destination information for the print job.

[0133] Next, in **S1309**, the cloud print server **107** executes a login processing to the cloud storage **106** by using the authentication information obtained in **S1308**.

[0134] Next, in **S1310**, the cloud storage **106** transmits a login processing result notification to the cloud print server **107**. Here, it is assumed that a login processing result notification indicating the success of the login processing has been transmitted.

[0135] Next, in **S1311**, the cloud print server **107** transmits, to the cloud storage **106**, a print job save request that includes the print job and the storage destination information for the print job.

[0136] Next, in **S1312**, the cloud storage **106** performs a file save processing. In the file save processing, the obtained print job is saved in the directory indicated by the obtained storage destination information. Next, in **S1313**, the cloud storage **106** transmits, to the cloud print server **107**, a save completion notification which indicates that saving of the print job has been completed.

It should be noted that in the case that the cloud print server **107** receives a save failure notification and a plurality of cloud storages have been set, the cloud print server **107** may perform the following processes. Specifically, the cloud print server **107** obtains authentication information for the cloud storage with the next priority order setting and the storage destination information, and returns to **S1309** to perform a login processing to the cloud storage with the next priority order setting. After the process of **S1313** is completed, the processes of **S1314** to **S1315**, which are the same as the above-described processes of **S416** to **S417**, are performed. The steps of **S1301** to **S1315** are the description of the procedure for saving the print job in the cloud storage **106**.

[0137] Next, the flow, in which the image forming apparatus **101** obtains print job related information from the cloud print server **107**, obtains a print job corresponding to the print job related information from the cloud storage **106**, and executes a print processing, will be described.

[0138] First, the processes of **S1316** to **S1325**, which are the same as the above-described processes of **S418** to **S427**, are performed.

[0139] Next, in **S1326**, the image forming apparatus **101** transmits, to the cloud print server **107**, a request for obtaining the print job that has been selected by the user in **S1323** (an obtaining request).

[0140] Next, in **S1327**, the cloud print server **107** obtains the authentication information for the cloud storage **106** that has saved the print job designated by the obtaining request, and the storage destination information.

[0141] Next, in **S1328**, the cloud print server **107** executes a login processing to the cloud storage **106** by using the authentication information obtained in **S1327**.

[0142] Next, in **S1329**, the cloud storage **106** transmits a login processing result notification to the cloud print server **107**. Here, it is assumed that a login processing result notification indicating the success of the login processing has been transmitted.

[0143] Next, in **S1330**, the cloud print server **107** transmits a request for obtaining the print job (an obtaining request) by using the storage destination information obtained in **S1327**.

[0144] Next, in **S1331**, the cloud storage **106** transmits a print job corresponding to this obtaining request to the cloud print server **107**.

[0145] In **S1332**, the cloud print server **107** transmits the print job that has been received from the cloud storage **106** to the image forming apparatus **101**.

[0146] In **S1333**, the image forming apparatus **101** executes a print processing of the print job that has been received from the cloud print server **107**. Thereafter, the third cloud print processing ends.

[0147] In the above-described third embodiment, the save location of the print job is selected by the cloud print server **107**. As a result, it is possible to reduce the user's labor and time in selecting

the save location of the print job.

[0148] In addition, in the above-described third embodiment, a first account for a user to use the cloud print server **107** has been linked to (has been associated with) a second account for the user to use the cloud storage **106**. The cloud print server **107** selects a cloud storage corresponding to the second account that has been linked to the first account of the user. As a result, by simply logging in to the cloud print server **107**, the user is able to set the cloud storage corresponding to the second account that has been linked to the first account of the user as the save location of the print job.

[0149] In addition, in the above-described embodiment, when logging in to the cloud print server **107** or the cloud storage **106**, a one-time password or the like may be set to improve security.

Other Embodiments

[0150] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium.

[0151] The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

[0152] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0153] This application claims the benefit of Japanese Patent Application No. 2024-029949, filed on Feb. 29, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A printing system that provides a cloud print service, the printing system comprising: at least one processor; and a memory coupled to the processor storing instructions that, when executed by the processor, cause the processor to perform causing an information processing apparatus, which has received a print job transmission instruction from a user, to generate a print job and transmit the print job to a save location that has been set; and causing an image forming apparatus, which has received a print execution instruction from the user, to obtain the print job that has been saved in the save location and execute a print processing of the obtained print job, and wherein the save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.
2. The printing system according to claim 1, wherein the save location is a cloud storage that has been selected from a plurality of cloud storages.
3. The printing system according to claim 2, further comprising: the cloud print server, and wherein

the save location is a cloud storage that has been selected by the cloud print server.

4. The printing system according to claim 3, wherein the processor is caused to further perform linking a first account for the user to use the cloud print server to a second account for the user to use the cloud storage, and the cloud print server selects a cloud storage corresponding to the second account that has been linked to the first account of the user.

5. The printing system according to claim 1, wherein the cloud print service is realized without using the cloud print server.

6. A control method for a printing system that provides a cloud print service, the control method comprising: a step of causing an information processing apparatus, which has received a print job transmission instruction from a user, to generate a print job and transmit the print job to a save location that has been set; and a step of causing an image forming apparatus, which has received a print execution instruction from the user, to obtain the print job that has been saved in the save location and execute a print processing of the obtained print job, and wherein the save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.

7. An information processing apparatus that causes an image forming apparatus, which supports a cloud print service, to execute a print processing, the information processing apparatus comprising: at least one processor; and a memory coupled to the processor storing instructions that, when executed by the processor, cause the processor to perform receiving an instruction from a user; and a control of generating a print job in accordance with the instruction and transmitting the print job to a save location that has been set, and wherein the image forming apparatus obtains the print job that has been saved in the save location in accordance with a print execution instruction received from the user, and executes a print processing of the obtained print job, and the save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.

8. A control method for an information processing apparatus that causes an image forming apparatus, which supports a cloud print service, to execute a print processing, the control method comprising: a step of receiving an instruction from a user; and a step of transmitting a print job to a save location that has been set in accordance with the instruction, and wherein the image forming apparatus obtains the print job that has been saved in the save location in accordance with a print execution instruction received from the user, and executes a print processing of the obtained print job, and the save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.

9. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a control method for an information processing apparatus that causes an image forming apparatus, which supports a cloud print service, to execute a print processing, the control method comprising: a step of receiving an instruction from a user; and a step of transmitting a print job to a save location that has been set in accordance with the instruction, and wherein the image forming apparatus obtains the print job that has been saved in the save location in accordance with a print execution instruction received from the user, and executes a print processing of the obtained print job, and the save location is a cloud storage other than a storage of a cloud print server managed by a manufacturer of the image forming apparatus.
